

DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

Sec:- 3

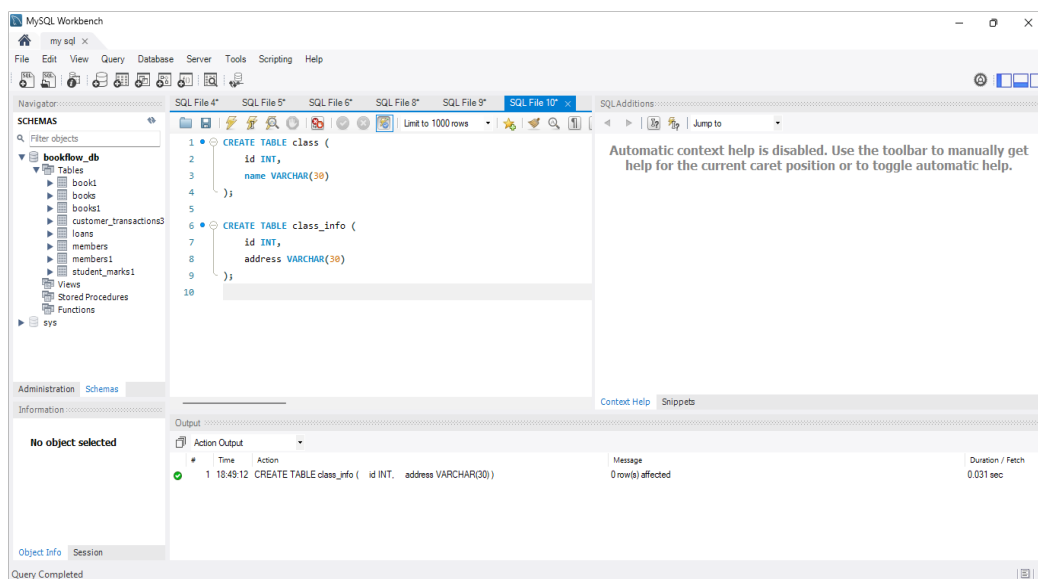
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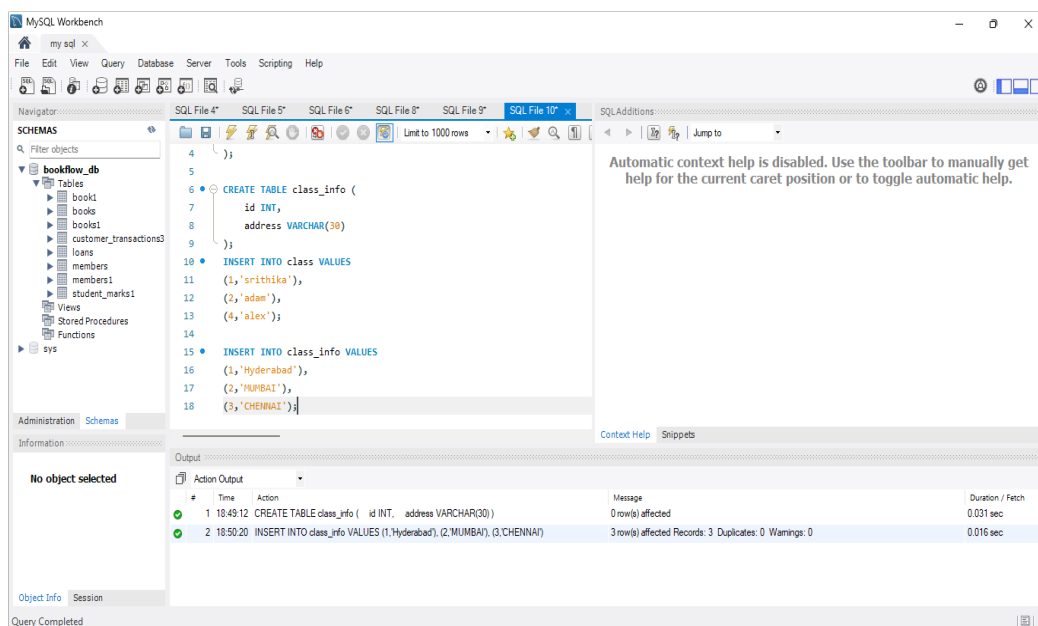
4 - Joint Set Operations

PART 1: CROSS JOIN

STEP 1: CREATE TABLES



STEP 2: INSERT DATA



QUESTION 1 - CROSS JOIN

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' schema with various tables and views. The main editor window contains the following SQL query:

```
20 INSERT INTO class_info VALUES
21 (1,'Hyderabad'),
22 (2,'MUMBAI'),
23 (3,'CHENNAI');
24 SELECT *
25 FROM class
26 CROSS JOIN class_info;
```

The 'Result Grid' shows the output of the query:

	id	name	id	address
1	alex	1	Hyderabad	
2	adam	1	Hyderabad	
3	srithika	1	Hyderabad	
4	alex	2	MUMBAI	
5	adam	2	MUMBAI	
6	srithika	2	MUMBAI	
7	alex	3	CHENNAI	
8	adam	3	CHENNAI	
9	srithika	3	CHENNAI	

The 'Output' pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
5	18:51:33	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	Error Code: 1146. Table 'bookflow_db.class' doesn't exist	0.000 sec
6	18:52:28	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	Error Code: 1146. Table 'bookflow_db.class' doesn't exist	0.000 sec
7	18:53:05	CREATE TABLE class (id INT, name VARCHAR(50))	0 row(s) affected	0.032 sec
8	18:53:28	INSERT INTO class VALUES (1,'srithika'), (2,'adam'), (4,'alex')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
9	18:53:48	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec

PART 2: INNER JOIN

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' schema. The main editor window contains the following SQL query:

```
34 address VARCHAR(30)
35 );
36
37 INSERT INTO class VALUES
38 (1,'srithika'),
39 (2,'adam'),
40 (3,'alex'),
41 (4,'anu');
42
43 INSERT INTO class_info VALUES
44 (1,'Hyderabad'),
45 (2,'MUMBAI'),
46 (3,'CHENNAI');
47
48
```

The 'Output' pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
6	18:52:28	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	Error Code: 1146. Table 'bookflow_db.class' doesn't exist	0.000 sec
7	18:53:05	CREATE TABLE class (id INT, name VARCHAR(50))	0 row(s) affected	0.032 sec
8	18:53:28	INSERT INTO class VALUES (1,'srithika'), (2,'adam'), (4,'alex')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
9	18:53:48	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
10	18:55:37	INSERT INTO class_info VALUES (1,'Hyderabad'), (2,'MUMBAI'), (3,'CHENNAI')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec

QUESTION 2 - INNER JOIN

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow_db

Tables

book1

books

books1

customer_transactions3

loans

members

members1

student_marks1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

Schema: bookflow_db

Object Info Session

SQL File 4* SQL File 5* SQL File 6* SQL File 8* SQL File 9* SQL File 10* x

Limit to 1000 rows

SQL

```
(3, 'CHENNAI');
46
47 • SELECT *
48 FROM class
49 INNER JOIN class_info
50 ON class.id = class_info.id;
51
52
```

Result Grid

	id	name	id	address
1	sritika	1	Hyderabad	
2	adam	2	MUMBAI	
1	sritika	1	Hyderabad	
2	adam	2	MUMBAI	

Result 2 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	18:53:05	CREATE TABLE class (id INT, name VARCHAR(50))	0 row(s) affected	0.032 sec
8	18:53:28	INSERT INTO class VALUES (1,'sritika'), (2,'adam'), (4,'alex')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
9	18:53:48	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
10	18:55:37	INSERT INTO class_info VALUES (1,Hyderabad), (2,MUMBAI), (3,CHENNAI)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
11	18:56:39	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec

Query Completed

QUESTION 3 - INNER JOIN

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow_db

Tables

book1

books

books1

customer_transactions3

loans

members

members1

student_marks1

Views

Stored Procedures

Functions

sys

Administration Schemas

Information

Schema: bookflow_db

Object Info Session

SQL File 4* SQL File 5* SQL File 6* SQL File 8* SQL File 9* SQL File 10* x

Limit to 1000 rows

SQL

```
50 ON class.id = class_info.id;
51 • SELECT class.name,
52 class_info.address
53 FROM class
54 INNER JOIN class_info
55 ON class.id=class_info.id;
56
```

Result Grid

	name	address
sritika	Hyderabad	
adam	MUMBAI	
sritika	Hyderabad	
adam	MUMBAI	

Result 3 x

Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	18:53:28	INSERT INTO class VALUES (1,'sritika'), (2,'adam'), (4,'alex')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
9	18:53:48	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
10	18:55:37	INSERT INTO class_info VALUES (1,Hyderabad), (2,MUMBAI), (3,CHENNAI)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
11	18:56:39	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
12	18:58:13	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class...	4 row(s) returned	0.000 sec / 0.000 sec

Query Completed

PART 3: NATURAL JOIN

QUESTION 4 - NATURAL JOIN

The screenshot shows the MySQL Workbench interface with a SQL query in the editor and its results in the Output pane.

SQL Query:

```
53 FROM class
54 INNER JOIN class_info
55 ON class.id=class_info.id;
56 SELECT *
57 FROM class
58 NATURAL JOIN class_info;
59
```

Result Grid:

id	name	address
1	srithika	Hyderabad
2	adam	MUMBAI
1	srithika	Hyderabad
2	adam	MUMBAI

Output:

#	Time	Action	Message	Duration / Fetch
9	18:53:48	SELECT * FROM class CROSS JOIN class_info LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
10	18:55:37	INSERT INTO class_info VALUES (1,Hyderabad), (2,MUMBAI), (3,CHENNAI)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
11	18:56:39	SELECT * FROM class INNER JOIN class_info ON class id = class_info id LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
12	18:58:13	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class...	4 row(s) returned	0.000 sec / 0.000 sec
13	18:59:19	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	4 row(s) returned	0.015 sec / 0.000 sec

PART 4: LEFT OUTER JOIN

The screenshot shows the MySQL Workbench interface with a SQL query in the editor and its results in the Output pane.

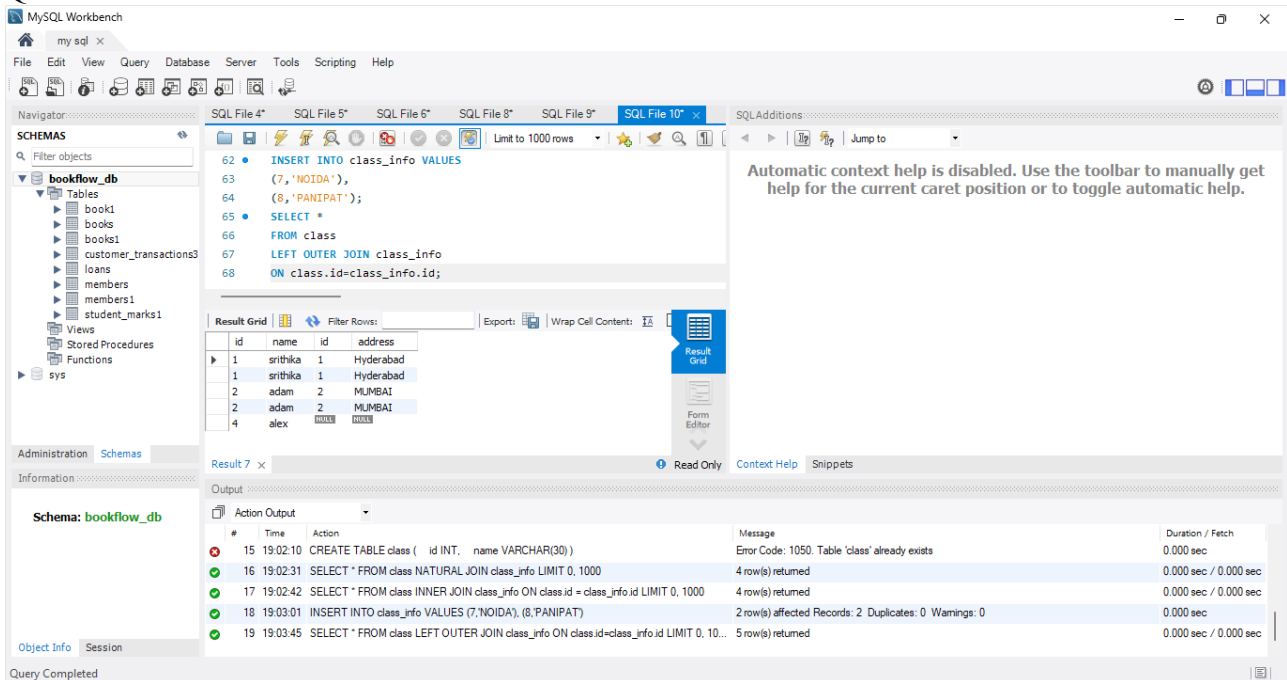
SQL Query:

```
52 class_info.address
53 FROM class
54 INNER JOIN class_info
55 ON class.id=class_info.id;
56 SELECT *
57 FROM class
58 NATURAL JOIN class_info;
59 INSERT INTO class VALUES
60 (5,'ashish');
61
62 INSERT INTO class_info VALUES
63 (7,'NOIDA'),
64 (8,'PANIPAT');
65
66
67
```

Output:

#	Time	Action	Message	Duration / Fetch
10	18:55:37	INSERT INTO class_info VALUES (1,Hyderabad), (2,MUMBAI), (3,CHENNAI)	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
11	18:56:39	SELECT * FROM class INNER JOIN class_info ON class id = class_info id LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
12	18:58:13	SELECT class.name, class_info.address FROM class INNER JOIN class_info ON class...	4 row(s) returned	0.000 sec / 0.000 sec
13	18:59:19	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	4 row(s) returned	0.015 sec / 0.000 sec
14	19:01:58	INSERT INTO class_info VALUES (7,NOIDA), (8,PANIPAT)	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.016 sec

QUESTION 5 - LEFT JOIN



MySQL Workbench interface showing a SQL query and its results. The query is a LEFT OUTER JOIN between 'class' and 'class_info' tables. The result grid shows 5 rows. The output pane shows the execution log.

SQL Query:

```
62 INSERT INTO class_info VALUES
63 (7,'NOIDA'),
64 (8,'PANIPAT');
65 SELECT *
66 FROM class
67 LEFT OUTER JOIN class_info
68 ON class.id=class_info.id;
```

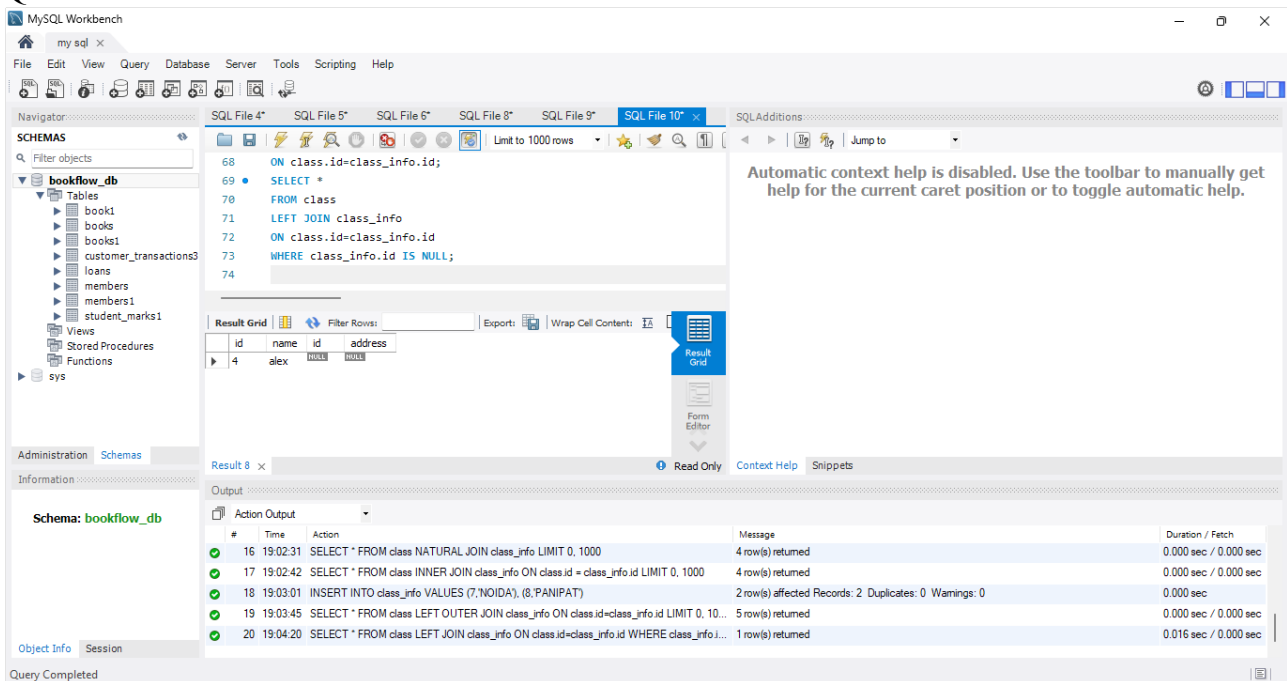
Result Grid:

	id	name	id	address
1	1	sritika	1	Hyderabad
1	1	sritika	1	Hyderabad
2	2	adam	2	MUMBAI
2	2	adam	2	MUMBAI
4	4	alex	NULL	NULL

Output:

#	Time	Action	Message	Duration / Fetch
15	19.02.10	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050, Table 'class' already exists	0.000 sec
16	19.02.31	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
17	19.02.42	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
18	19.03.01	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
19	19.03.45	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id=class_info.id LIMIT 0, 10...	5 row(s) returned	0.000 sec / 0.000 sec

QUESTION 6 - LEFT JOIN



MySQL Workbench interface showing a SQL query and its results. The query is a LEFT JOIN between 'class' and 'class_info' tables with a WHERE clause. The result grid shows 1 row. The output pane shows the execution log.

SQL Query:

```
68 ON class.id=class_info.id;
69 SELECT *
70 FROM class
71 LEFT JOIN class_info
72 ON class.id=class_info.id
73 WHERE class_info.id IS NULL;
74
```

Result Grid:

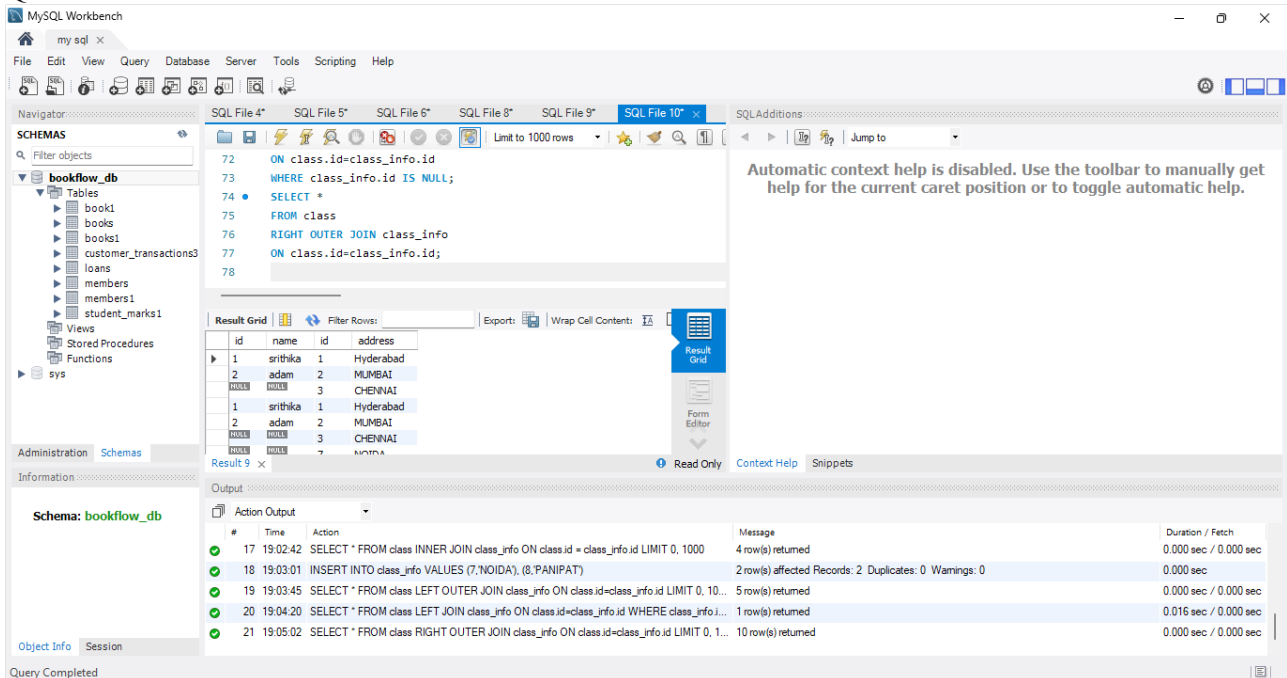
	id	name	id	address
4	4	alex	NULL	NULL

Output:

#	Time	Action	Message	Duration / Fetch
16	19.02.31	SELECT * FROM class NATURAL JOIN class_info LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
17	19.02.42	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
18	19.03.01	INSERT INTO class_info VALUES (7,'NOIDA'), (8,'PANIPAT')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
19	19.03.45	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id=class_info.id LIMIT 0, 10...	5 row(s) returned	0.000 sec / 0.000 sec
20	19.04.20	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id WHERE class_info...	1 row(s) returned	0.016 sec / 0.000 sec

PART 5: RIGHT OUTER JOIN

QUESTION 7 - RIGHT JOIN



MySQL Workbench interface showing a SQL query for a RIGHT OUTER JOIN. The query is as follows:

```
72 ON class.id=class_info.id
73 WHERE class_info.id IS NULL;
74 SELECT *
75 FROM class
76 RIGHT OUTER JOIN class_info
77 ON class.id=class_info.id;
78
```

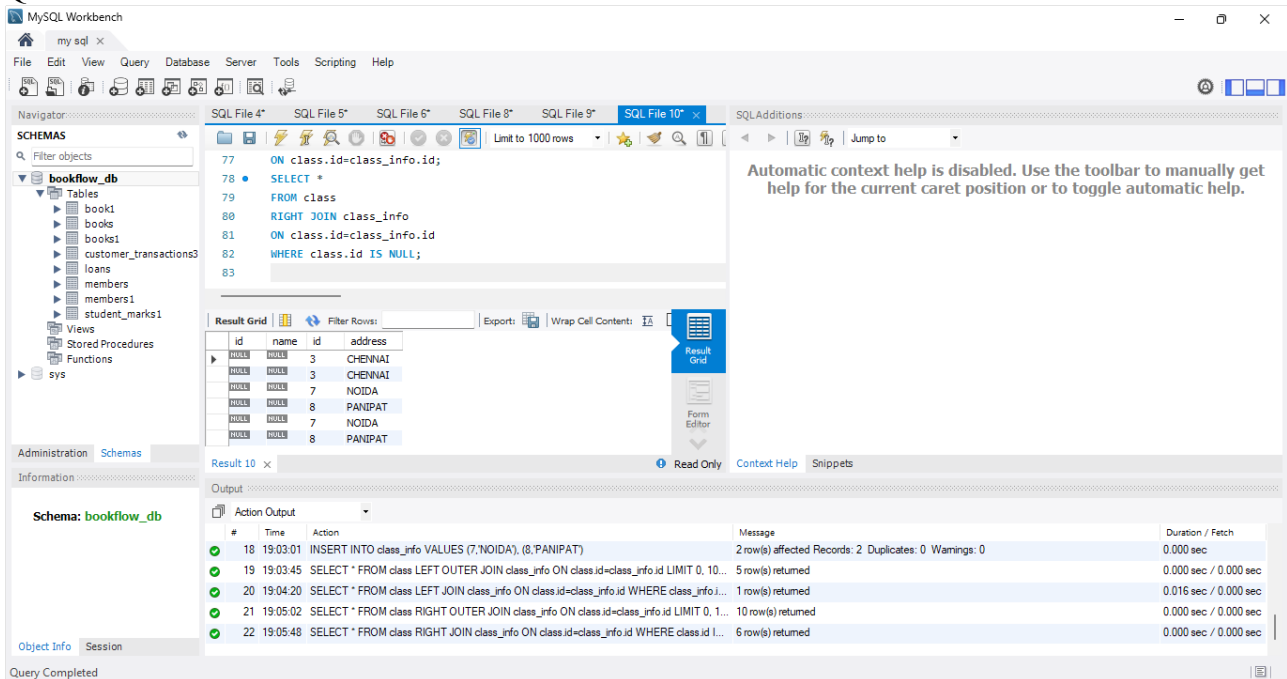
The Result Grid shows the following data:

	id	name	id	address
1	1	srithika	1	Hyderabad
2	2	adam	2	MUMBAI
3	3	NOIDA	3	CHENNAI
4	1	srithika	1	Hyderabad
5	2	adam	2	MUMBAI
6	3	NOIDA	3	CHENNAI
7	3	CHENNAI	3	CHENNAI

The Output pane shows the following actions:

#	Time	Action	Message	Duration / Fetch
17	19:02:42	SELECT * FROM class INNER JOIN class_info ON class.id = class_info.id LIMIT 0, 1000	4 row(s) returned	0.000 sec / 0.000 sec
18	19:03:01	INSERT INTO class_info VALUES (7,NOIDA), (8,PANIPAT)	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
19	19:03:45	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id=class_info.id LIMIT 0, 10...	5 row(s) returned	0.000 sec / 0.000 sec
20	19:04:20	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id WHERE class_info.i...	1 row(s) returned	0.016 sec / 0.000 sec
21	19:05:02	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id=class_info.id LIMIT 0, 1...	10 row(s) returned	0.000 sec / 0.000 sec

QUESTION 8 - RIGHT JOIN



MySQL Workbench interface showing a SQL query for a RIGHT JOIN. The query is as follows:

```
77 ON class.id=class_info.id;
78 SELECT *
79 FROM class
80 RIGHT JOIN class_info
81 ON class.id=class_info.id
82 WHERE class.id IS NULL;
83
```

The Result Grid shows the following data:

	id	name	id	address
1	3	CHENNAI	3	CHENNAI
2	7	NOIDA	7	NOIDA
3	8	PANIPAT	8	PANIPAT
4	7	NOIDA	7	NOIDA
5	8	PANIPAT	8	PANIPAT

The Output pane shows the following actions:

#	Time	Action	Message	Duration / Fetch
18	19:03:01	INSERT INTO class_info VALUES (7,NOIDA), (8,PANIPAT)	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.000 sec
19	19:03:45	SELECT * FROM class LEFT OUTER JOIN class_info ON class.id=class_info.id LIMIT 0, 10...	5 row(s) returned	0.000 sec / 0.000 sec
20	19:04:20	SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id WHERE class_info.i...	1 row(s) returned	0.016 sec / 0.000 sec
21	19:05:02	SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id=class_info.id LIMIT 0, 1...	10 row(s) returned	0.000 sec / 0.000 sec
22	19:05:48	SELECT * FROM class RIGHT JOIN class_info ON class.id=class_info.id WHERE class.id I...	6 row(s) returned	0.000 sec / 0.000 sec

PART 6: FULL OUTER JOIN

QUESTION 9 - FULL OUTER JOIN

The screenshot shows MySQL Workbench with a query in SQL File 10. The query is as follows:

```
81 ON class.id=class_info.id
82 WHERE class.id IS NULL;
83 • SELECT * FROM class
84 LEFT JOIN class_info ON class.id = class_info.id
85 UNION
86 SELECT * FROM class
87 RIGHT JOIN class_info ON class.id = class_info.id;
```

The Result Grid shows the following data:

id	name	id	address
1	srithika	1	Hyderabad
2	adam	2	MUMBAI
4	alex	NULL	NULL
NULL	NULL	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

The Output pane shows the execution log with the following messages:

- 20 19:04:20 SELECT * FROM class LEFT JOIN class_info ON class.id=class_info.id WHERE class_info.id... 1 row(s) returned 0.016 sec / 0.000 sec
- 21 19:05:02 SELECT * FROM class RIGHT OUTER JOIN class_info ON class.id=class_info.id LIMIT 0, 1... 10 row(s) returned 0.000 sec / 0.000 sec
- 22 19:05:48 SELECT * FROM class RIGHT JOIN class_info ON class.id=class_info.id WHERE class.id l... 6 row(s) returned 0.000 sec / 0.000 sec
- 23 19:07:02 SELECT * FROM class FULL OUTER JOIN class_info ON class.id=class_info id Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ... 0.016 sec
- 24 19:08:07 SELECT * FROM class LEFT JOIN class_info ON class.id = class_info id UNION SELECT * ... 6 row(s) returned 0.016 sec / 0.000 sec

QUESTION 10 - FULL OUTER JOIN

The screenshot shows MySQL Workbench with a query in SQL File 10. The query is as follows:

```
88 • SELECT * FROM class
89 LEFT JOIN class_info ON class.id = class_info.id
90 WHERE class_info.id IS NULL
91 UNION
92 SELECT * FROM class
93 RIGHT JOIN class_info ON class.id = class_info.id
94 WHERE class.id IS NULL;
```

The Result Grid shows the following data:

id	name	id	address
4	alex	NULL	NULL
NULL	NULL	3	CHENNAI
NULL	NULL	7	NOIDA
NULL	NULL	8	PANIPAT

The Output pane shows the execution log with the following messages:

- 22 19:05:48 SELECT * FROM class RIGHT JOIN class_info ON class.id=class_info id WHERE class id l... 6 row(s) returned 0.000 sec / 0.000 sec
- 23 19:07:02 SELECT * FROM class FULL OUTER JOIN class_info ON class id=class_info id Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ... 0.016 sec
- 24 19:08:07 SELECT * FROM class LEFT JOIN class_info ON class.id = class_info id UNION SELECT * ... 6 row(s) returned 0.016 sec / 0.000 sec
- 25 19:08:59 SELECT * FROM class FULL OUTER JOIN class_info ON class id=class_info id WHERE cl... Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ... 0.000 sec
- 26 19:09:34 SELECT * FROM class LEFT JOIN class_info ON class.id = class_info id WHERE class_inf... 4 row(s) returned 0.000 sec / 0.000 sec

PART 7: UNION

MySQL Workbench interface showing the following SQL queries in the editor:

```
105 id INT PRIMARY KEY,  
106 name VARCHAR(50)  
107 );  
108  
109  
110 • INSERT INTO first_table VALUES  
111 (1,'abhi'),  
112 (2,'adam');  
113  
114 • INSERT INTO second_table VALUES  
115 (2,'adam'),  
116 (3,'chester');  
117  
118  
119  
120
```

The Output window displays the following results:

#	Time	Action	Message	Duration / Fetch
25	19:08:59	SELECT * FROM class FULL OUTER JOIN class_info ON class_id=class_info.id WHERE cl...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ...	0.000 sec
26	19:09:34	SELECT * FROM class LEFT JOIN class_info ON class_id = class_info.id WHERE class_inf...	4 row(s) returned	0.000 sec / 0.000 sec
27	10:51:21	INSERT INTO second_table VALUES (2,'adam'), (3,'chester')	Error Code: 1146. Table 'bookflow_db.second_table' doesn't exist	0.016 sec
28	10:53:20	CREATE TABLE second_table (id INT PRIMARY KEY, name VARCHAR(50))	0 row(s) affected	0.047 sec
29	10:53:26	INSERT INTO second_table VALUES (2,'adam'), (3,'chester')	2 row(s) affected Records: 2 Duplicates: 0 Warnings: 0	0.016 sec

QUESTION 11 – UNION

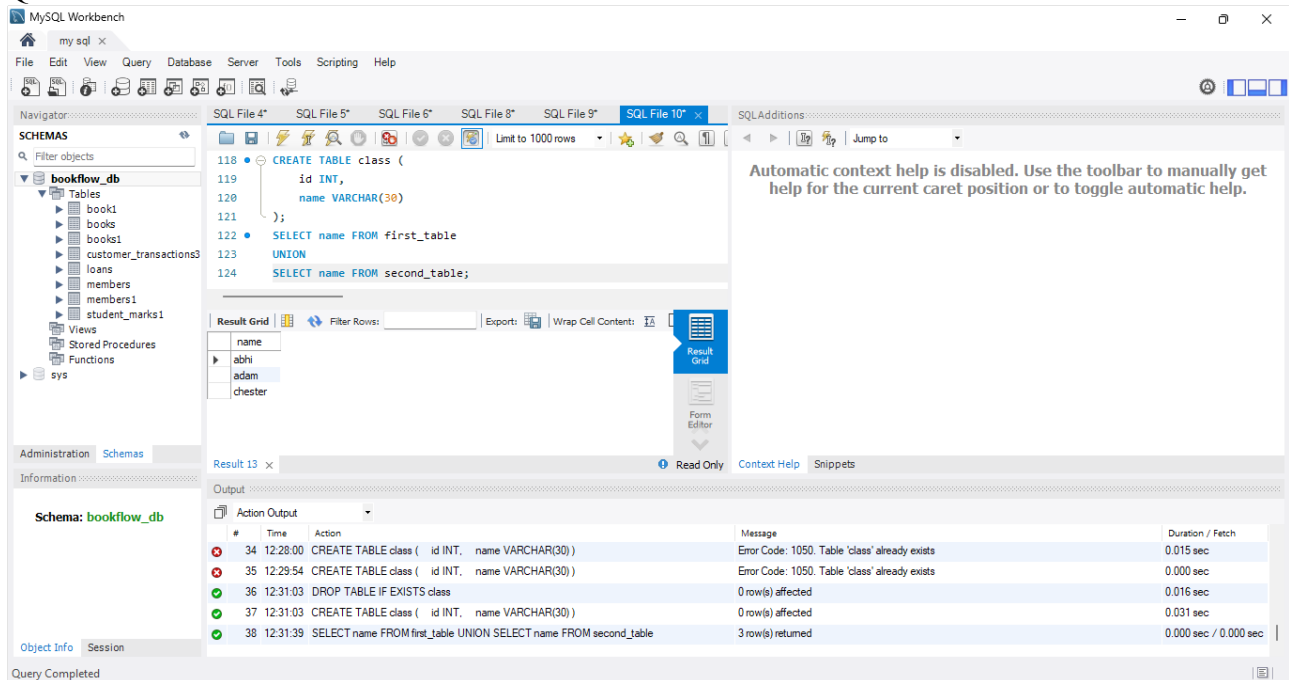
MySQL Workbench interface showing the following SQL queries in the editor:

```
107 );  
108  
109  
110 • INSERT INTO first_table VALUES  
111 (1,'abhi'),  
112 (2,'adam');  
113  
114 • INSERT INTO second_table VALUES  
115 (2,'adam'),  
116 (3,'chester');  
117  
118 • DROP TABLE IF EXISTS class;  
119  
120 CREATE TABLE class (  
121 id INT,  
122 name VARCHAR(30)  
123 );  
124
```

The Output window displays the following results:

#	Time	Action	Message	Duration / Fetch
33	12:18:28	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050. Table 'class' already exists	0.000 sec
34	12:28:00	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050. Table 'class' already exists	0.015 sec
35	12:29:54	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050. Table 'class' already exists	0.000 sec
36	12:31:03	DROP TABLE IF EXISTS class	0 row(s) affected	0.016 sec
37	12:31:03	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected	0.031 sec

QUESTION 12 – UNION



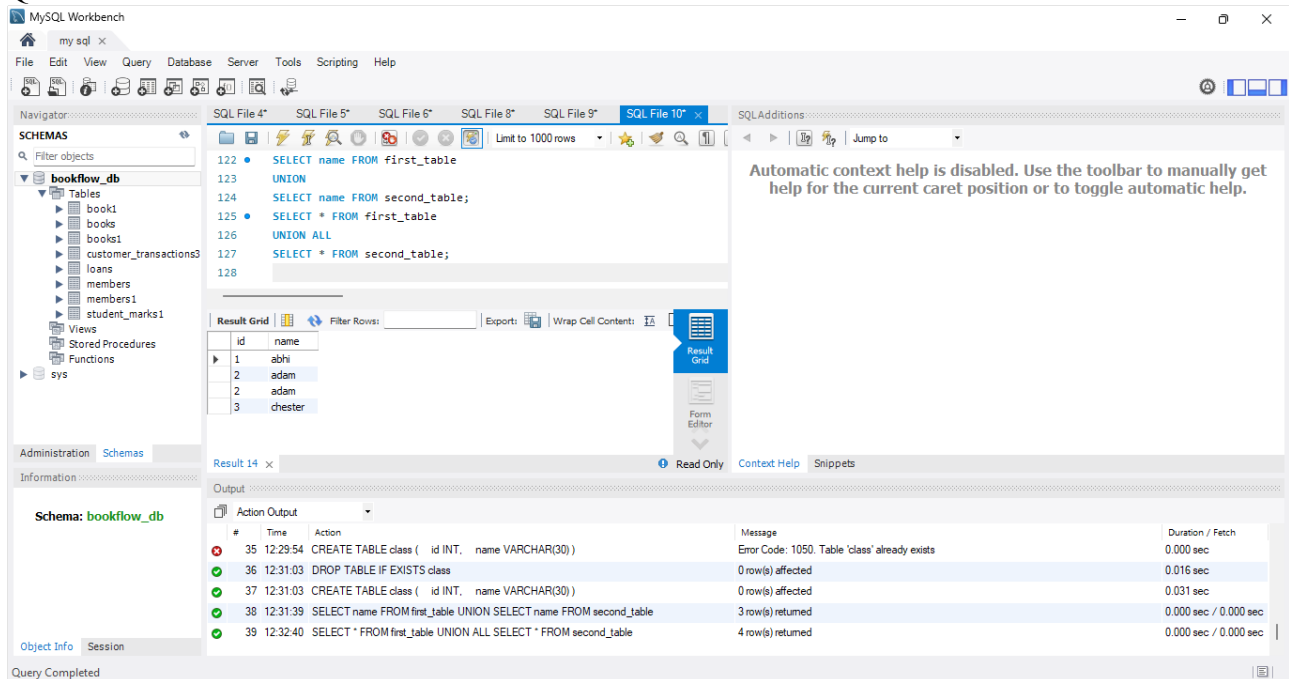
Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
118 CREATE TABLE class (  
119     id INT,  
120     name VARCHAR(30)  
121 );  
122 SELECT name FROM first_table  
123 UNION  
124 SELECT name FROM second_table;
```

#	Time	Action	Message	Duration / Fetch
34	12:28:00	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050, Table 'class' already exists	0.015 sec
35	12:29:54	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050, Table 'class' already exists	0.000 sec
36	12:31:03	DROP TABLE IF EXISTS class	0 row(s) affected	0.016 sec
37	12:31:03	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected	0.031 sec
38	12:31:39	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec

PART 8: UNION ALL

QUESTION 13 - UNION ALL



Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```
122 SELECT name FROM first_table  
123 UNION  
124 SELECT name FROM second_table;  
125 SELECT * FROM first_table  
126 UNION ALL  
127 SELECT * FROM second_table;  
128
```

#	Time	Action	Message	Duration / Fetch
35	12:29:54	CREATE TABLE class (id INT, name VARCHAR(30))	Error Code: 1050, Table 'class' already exists	0.000 sec
36	12:31:03	DROP TABLE IF EXISTS class	0 row(s) affected	0.016 sec
37	12:31:03	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected	0.031 sec
38	12:31:39	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
39	12:32:40	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec

QUESTION 14 - UNION ALL

The screenshot shows MySQL Workbench with a query editor containing the following SQL code:

```
129 FROM
130 (
131   SELECT * FROM first_table
132   UNION ALL
133   SELECT * FROM second_table
134 ) A;
135
```

The **Result Grid** shows the following data:

COUNT(*)
4

The **Output** pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
36	12:31:03	DROP TABLE IF EXISTS class	0 row(s) affected	0.016 sec
37	12:31:03	CREATE TABLE class (id INT, name VARCHAR(30))	0 row(s) affected	0.031 sec
38	12:31:39	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
39	12:32:40	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec
40	14:28:23	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM seco...	1 row(s) returned	0.000 sec / 0.000 sec

Query Completed

PART 9: INTERSECT

QUESTION 15 – INTERSECT

The screenshot shows MySQL Workbench with a query editor containing the following SQL code:

```
131 SELECT * FROM first_table
132 UNION ALL
133 SELECT * FROM second_table
134 ) A;
135 SELECT * FROM first_table
136 INTERSECT
137 SELECT * FROM second_table;
```

The **Result Grid** shows the following data:

id	name
2	adam

The **Output** pane shows the execution log:

#	Time	Action	Message	Duration / Fetch
38	12:31:39	SELECT name FROM first_table UNION SELECT name FROM second_table	3 row(s) returned	0.000 sec / 0.000 sec
39	12:32:40	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec
40	14:28:23	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM seco...	1 row(s) returned	0.000 sec / 0.000 sec
41	14:29:26	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned	0.016 sec / 0.000 sec
42	14:29:37	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec

Query Completed

QUESTION 16 – INTERSECT

The screenshot shows MySQL Workbench with a query window containing the following SQL code:

```
135 SELECT * FROM first_table;
136 INTERSECT
137 SELECT * FROM second_table;
138 SELECT name FROM first_table;
139 INTERSECT
140 SELECT name FROM second_table;
141
```

The result grid displays one row with the name 'adam'. The output pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
39	12:32:40	SELECT * FROM first_table UNION ALL SELECT * FROM second_table	4 row(s) returned	0.000 sec / 0.000 sec
40	14:28:23	SELECT COUNT(*) FROM (SELECT * FROM first_table UNION ALL SELECT * FROM second_table)	1 row(s) returned	0.000 sec / 0.000 sec
41	14:29:26	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned	0.016 sec / 0.000 sec
42	14:29:37	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec
43	14:30:21	SELECT name FROM first_table INTERSECT SELECT name FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec

PART 10: MINUS

QUESTION 17 – MINUS

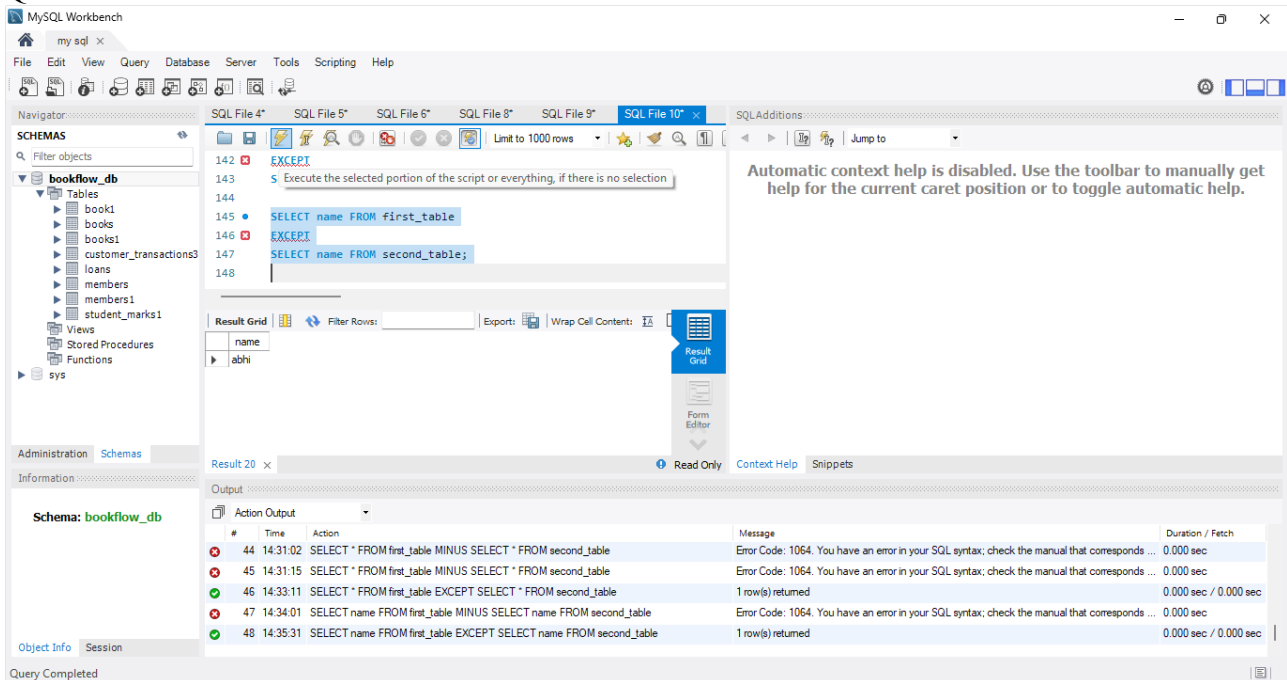
The screenshot shows MySQL Workbench with a query window containing the following SQL code:

```
137 SELECT * FROM second_table;
138 SELECT name FROM first_table;
139 INTERSECT
140 SELECT name FROM second_table;
141 SELECT * FROM first_table;
142 EXCEPT
143 SELECT * FROM second_table;
144
```

The result grid displays one row with the name 'abhi'. The output pane shows the execution log with the following entries:

#	Time	Action	Message	Duration / Fetch
42	14:29:37	SELECT * FROM first_table INTERSECT SELECT * FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec
43	14:30:21	SELECT name FROM first_table INTERSECT SELECT name FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec
44	14:31:02	SELECT * FROM first_table MINUS SELECT * FROM second_table	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to ...	0.000 sec
45	14:31:15	SELECT * FROM first_table MINUS SELECT * FROM second_table	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds to ...	0.000 sec
46	14:33:11	SELECT * FROM first_table EXCEPT SELECT * FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec

QUESTION 18 – MINUS



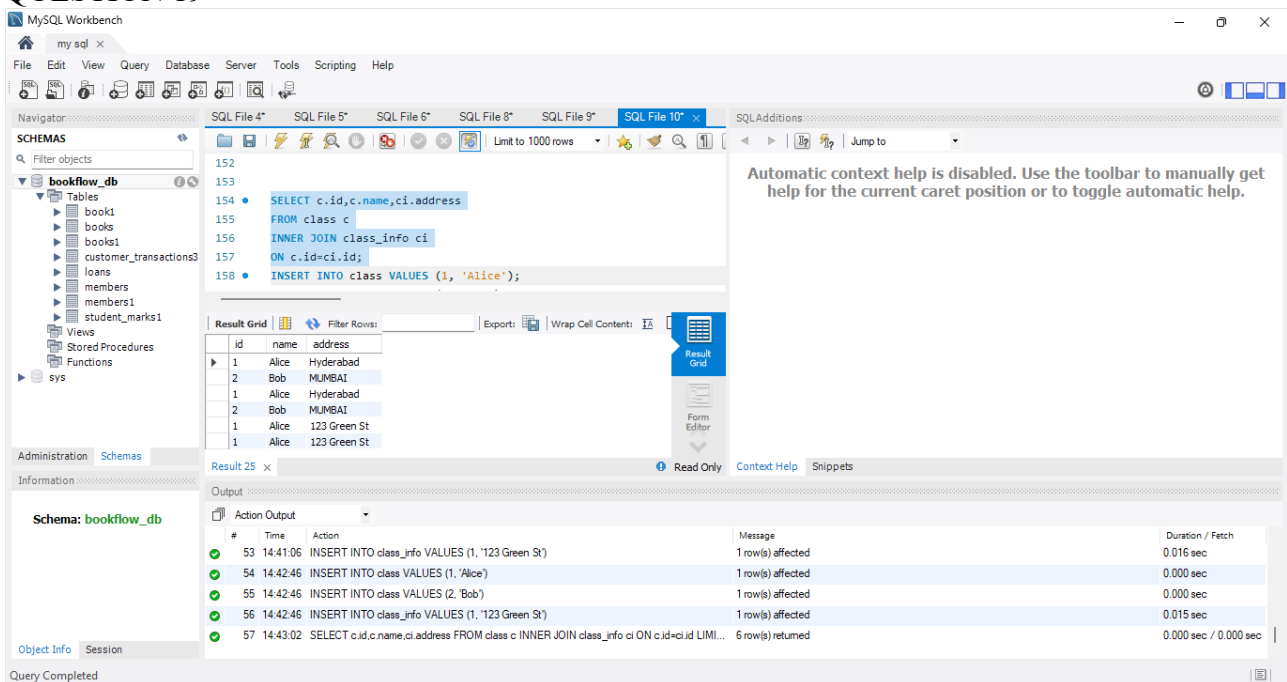
The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' schema with tables like 'book1', 'books', 'books1', 'customer_transactions3', 'loans', 'members', 'members1', 'student_marks1', 'Views', 'Stored Procedures', and 'Functions'. The main editor window shows a SQL script with a 'MINUS' operation. The 'Result Grid' shows a single row with the name 'abhi'. The 'Output' pane at the bottom shows the execution log with error messages for the 'MINUS' operation.

```
142 EXCEPT
143 S Execute the selected portion of the script or everything, if there is no selection
144
145 SELECT name FROM first_table
146 EXCEPT
147 SELECT name FROM second_table;
148
```

#	Time	Action	Message	Duration / Fetch
44	14:31:02	SELECT * FROM first_table MINUS SELECT * FROM second_table	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ...	0.000 sec
45	14:31:15	SELECT * FROM first_table MINUS SELECT * FROM second_table	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ...	0.000 sec
46	14:33:11	SELECT * FROM first_table EXCEPT SELECT * FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec
47	14:34:01	SELECT name FROM first_table MINUS SELECT name FROM second_table	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ...	0.000 sec
48	14:35:31	SELECT name FROM first_table EXCEPT SELECT name FROM second_table	1 row(s) returned	0.000 sec / 0.000 sec

ADVANCED INTERVIEW QUESTIONS

QUESTION 19



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'bookflow_db' schema. The main editor window shows a SQL script with a complex query involving 'INNER JOIN' and 'VALUES'. The 'Result Grid' shows a table with columns 'id', 'name', and 'address'. The 'Output' pane at the bottom shows the execution log with messages for the 'INSERT INTO' and 'SELECT' operations.

```
152
153
154 SELECT c.id,c.name,ci.address
155 FROM class c
156 INNER JOIN class_info ci
157 ON c.id=ci.id;
158 INSERT INTO class VALUES (1, 'Alice');
```

#	Time	Action	Message	Duration / Fetch
53	14:41:06	INSERT INTO class_info VALUES (1, '123 Green St')	1 row(s) affected	0.016 sec
54	14:42:46	INSERT INTO class VALUES (1, 'Alice')	1 row(s) affected	0.000 sec
55	14:42:46	INSERT INTO class VALUES (2, 'Bob')	1 row(s) affected	0.000 sec
56	14:42:46	INSERT INTO class_info VALUES (1, '123 Green St')	1 row(s) affected	0.015 sec
57	14:43:02	SELECT c.id,c.name,ci.address FROM class c INNER JOIN class_info ci ON c.id=ci.id LIMIT ...	6 row(s) returned	0.000 sec / 0.000 sec

QUESTION 20

MySQL Workbench

my sql x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

bookflow_db

- Tables
 - book1
 - books
 - books1
 - customer_transactions3
 - loans
 - members
 - members1
 - student_marks1
- Views
- Stored Procedures
- Functions
- sys

SQL File 4* SQL File 5* SQL File 6* SQL File 8* SQL File 9* SQL File 10* x

Limit to 1000 rows

SQL File 10*

```
166 THEN 'Address Missing'
167 ELSE 'Address Available'
168 END AS Status
169 FROM class c
170 LEFT JOIN class_info ci
171 ON c.id=ci.id;
172
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

	id	name	Status
1	abhi	Address Available	
2	adam	Address Available	
3	alex	Address Available	
4	anu	Address Missing	
5	ashish	Address Missing	

Result 27 x

Read Only

Context Help

Snippets

Schema: bookflow_db

Object Info

Session

Query Completed

Action Output

#	Time	Action	Message	Duration / Fetch
66	14:45:20	INSERT INTO class_info VALUES (1, 'Street 1')	1 row(s) affected	0.015 sec
67	14:45:20	INSERT INTO class_info VALUES (2, 'Street 2')	1 row(s) affected	0.000 sec
68	14:45:20	INSERT INTO class_info VALUES (3, 'Street 3')	1 row(s) affected	0.000 sec
69	14:45:53	WHEN ci.address IS NULL THEN 'Address Missing' ELSE 'Address Available' ...	Error Code: 1064. You have an error in your SQL syntax; check the manual that corresponds ...	0.000 sec
70	14:46:31	SELECT c.id, c.name, CASE WHEN ci.address IS NULL THEN 'Addr...	5 row(s) returned	0.000 sec / 0.000 sec